

PRACTICE WORKSHOP

Workshop 6: Arrays of Objects & Collections

This workshop ties everything together: managing groups of objects using arrays and the Java Collections Framework (List, Set, Map), including searching, sorting, and simple aggregation.

Learning Objectives

- Create and manipulate an array of custom objects.
- Choose between List, Set, and Map based on the problem's requirements.
- Search, filter, and aggregate data held in a collection.
- Iterate collections safely using for-each and Iterator.

Exercises

Exercise 1: Class Roster (Array)

Using an array of a Student class (name, score), write methods to find the student with the highest score, compute the class average, and print all students sorted by score descending.

Hints:

- Arrays.sort with a custom Comparator can sort an object array.
- Track the max as you iterate rather than sorting first, for practice.

Exercise 2: Unique Tags (Set)

Given a list of article tag strings that may contain duplicates and inconsistent casing (e.g. 'Java', 'java', 'JAVA'), build a HashSet of normalized, unique tags and print how many duplicates were removed.

Hints:

- Normalize with toLowerCase() before adding to the set.
- Compare the size of the input list to the resulting set size.

Exercise 3: Inventory Lookup (Map)

Model a small shop's inventory as a `HashMap` mapping product name to quantity in stock. Write methods to restock a product, sell a product (rejecting the sale if there isn't enough stock), and print a low-stock report for any product below 5 units.

Hints:

- `getOrDefault()` is handy when a product name is not yet a key.
- Update quantities with `put()` after computing the new value.

```
import java.util.HashMap;
import java.util.Map;

public class Inventory {
    private Map stock = new HashMap<>();

    public void restock(String product, int amount) {
        // TODO
    }

    public boolean sell(String product, int amount) {
        // TODO: return false if not enough stock
    }

    public void printLowStockReport(int threshold) {
        // TODO
    }
}
```

Stretch Goal (optional)

Combine ideas from all three exercises: store `Student` objects in a `Map` keyed by student ID, and write a method that returns a `List` of everyone above the class average, sorted by name.

This worksheet was written as original practice material for the PRO192 course export and is not affiliated with or copied from any institution's official course content.